Correlated Country Risk and International Capital Allocation

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Abstract

International investors view neighboring countries as a package: they favor or avoid them together. We show that this neighborhood structure is an important force in the global allocation of capital, distinct from the familiar cost of investing far from home. We develop a multi-country general equilibrium model of portfolio allocation that separates the two. We recover destination attractiveness from bilateral portfolio holdings and identify its spatial component using the distance-dependent covariance across countries. It accounts for two-thirds of persistent cross-country differences in attractiveness. Sovereign credit risk and geopolitical alignment share its spatial structure, pointing to geographic patterns in risk and bloc formation as its sources. Together, the two channels account for 64 percent of the cross-country variance of log capital and 25 percent of that of log output, more than the sum of their separate contributions. The neighborhood component also governs how shocks travel. When confidence in one country falls, capital flees to distant economies and lowers their interest rates, but rates in its neighbors rise. Geography links destinations to one another, not only investors to destinations.

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1 Introduction

Although transaction and technological costs of moving capital across borders have fallen substantially, capital is not allocated evenly worldwide. Early work has shown that geographic distance is one factor that influences international investment position. Investors disproportionately hold domestic and nearby foreign assets; bilateral portfolio holdings decline with distance. These patterns are commonly interpreted as information frictions, familiarity, or other barriers to international investment.

However, geography may matter for international capital allocation in another way. Neighboring countries may also share characteristics, or global investors may view them similarly. While bilateral portfolio frictions depend on the investor country's location, geography may also shape international capital allocation independently of the investor country's location.

In this article, we study how these two geographic forces affect international capital allocation and cross-country interest rate differences. We first document that bilateral interest rate differentials widen as distance between two countries increases. The relationship holds even after we isolate persistent country-pair interest rate differences by removing country-time-specific shocks. Thus, the geographic relationship in interest rates is not only a transitory macroeconomic phenomenon but also a persistent feature of cross-country interest rate differences.

We distinguish two channels through which geography influences portfolio holdings. First, it operates between an investor and a destination. Investors face bilateral portfolio frictions that increase with geographic distance. They allocate less capital to destinations that are farther away. This may be because investors have less access to information about destination countries. For example, investors who investigate investment opportunities face higher costs when the destination is farther away.

Second, geography operates across destinations themselves. Destination-specific characteristics that affect investment demand may be spatially correlated, so that geographically nearby countries have similar investment attractiveness to investors from all origins. The second channel is distinct from the first and economically important. For instance, a U.S. investor may allocate less capital to Korea than to Canada because Korea is farther away from the United States. This is a bilateral investor-destination friction. At the same time, Korea may have a demand component similar to that of nearby countries such as Japan. This second geographic channel does not depend on the investor's location. U.S. and Australian investors alike may view Korea and Japan as having similar

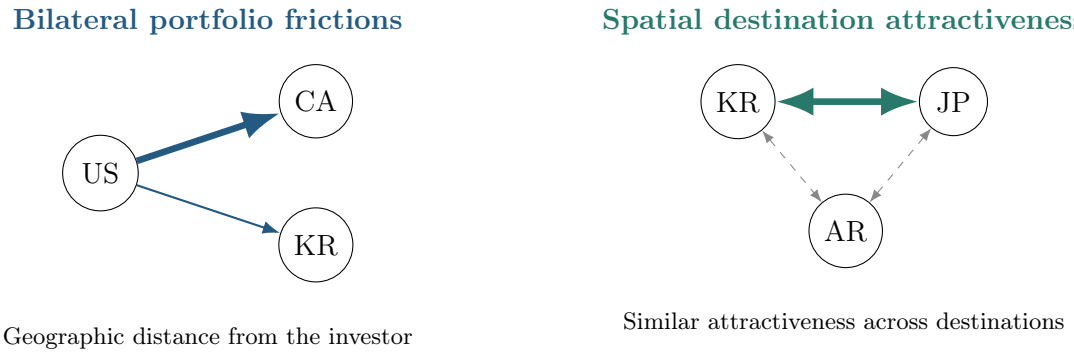


Figure 1: Geography Matters Through Two Distinct Channels

demand components.

To quantify these channels, we develop a multi-country production economy with international portfolio allocation. Financial intermediaries allocate households' savings across domestic and foreign assets. They make portfolio choices based on destination-country returns, a bilateral portfolio wedge, and a destination-specific attractiveness component not captured by returns. On the production side, countries have production technologies with decreasing returns to capital. In general equilibrium, international portfolio holdings and equilibrium interest rates are determined by market clearing.

Destination-specific attractiveness is linked to equilibrium interest rates. Higher attractiveness raises investment demand and capital inflows, increasing the capital stock and lowering interest rates. Consequently, when destination attractiveness contains a spatially correlated component, geographically proximate countries tend to exhibit more similar interest rates.

We estimate the bilateral portfolio wedge using international portfolio holdings data. The logit asset demand model allows us to decompose the portfolio equation into a bilateral wedge, a destination component, and an investor component, which we estimate using a parameterized wedge, investor-time fixed effects, and destination-time fixed effects. We find that portfolio holdings decline with geographic distance and are lower across geopolitical blocs.

Because interest rates are determined in equilibrium, we estimate the elasticity of portfolio demand using an instrumental variable strategy that builds on Koijen and Yogo (2026). The strategy uses variation in relative demand pressure and supply. We construct predicted demand using a third-party country's demand component implied by the estimated bilateral gravity forces, and predicted supply based on observable characteristics. Variation in relative demand pressure shifts equilibrium returns

through market clearing and identifies the elasticity of asset demand.

We next study the destination-side channel. We recover destination-specific attractiveness components and show that geographically proximate countries tend to have more similar attractiveness components. This relationship holds after removing market size effects or controlling for geopolitical distance. We then decompose the attractiveness components into a persistent spatial component, a persistent non-spatial component, and a transitory component. To capture spatial correlation, we model the covariance structure as a decreasing function of bilateral distance and estimate it using restricted maximum likelihood. The estimates show substantial spatial dependence in destination attractiveness.

This relationship has additional implications. Spatially correlated shocks can generate common financial and macroeconomic movements within a regional bloc that differ from those in other regions. A negative attractiveness shock drives capital outflows from the affected country, raising interest rates there while lowering them in unaffected countries. When the shock is spatially correlated, neighboring countries experience a decline in attractiveness, and their interest rates rise rather than fall, generating a regional response to the shock. Therefore, spatial dependence in destination attractiveness can change not only the magnitude of capital flow and interest rate responses but also their direction.

We use the estimated model to quantify the implications of two geographic channels. First, holding the investor-country asset fixed, we conduct a counterfactual analysis by removing bilateral frictions and the persistent spatial component of destination attractiveness. We compare interest rate differences relative to the U.S. The finding is that spatially correlated destination attractiveness accounts for most of the aggregate geographic contribution to interest rate differentials. However, there is heterogeneity across country groups behind this result. Geographic forces reduce interest rate differences relative to the U.S. among advanced countries, while increasing them in emerging market economies.

Second, we examine how spatially correlated shocks can drive regional interest rate movements that differ from those in other regions. Using the estimated spatial dependence in destination attractiveness, we compare negative idiosyncratic shocks with spatially correlated shocks that also affect neighboring countries. In contrast to idiosyncratic shocks, a spatially correlated negative shock induces capital outflows and higher interest rates in both the affected country and its neighbors,

while lowering interest rates in more distant countries.

Third, we allow savings, capital stocks, and equilibrium interest rates to adjust to a new steady state in the counterfactual economy. Removing either bilateral geographic frictions or the spatial component of destination attractiveness reduces international dispersion in capital and output, and removing both channels leads to larger reductions. The results show that both geographic channels play quantitatively important roles in capital allocation and output.

Our findings show that geography plays a significant role in international finance, and its role is not solely characterized by bilateral portfolio frictions. Geographic distance also shapes the relationship between investment destinations, creating spatial dependence that affects general equilibrium outcomes. The two channels operate through distinct mechanisms, and both have quantitative implications.

Our paper contributes to the literature in several ways. First, we connect the geographic patterns in international portfolio holdings to interest rate differentials and general equilibrium models. Second, we distinguish spatial dependence across destinations from bilateral investor-destination frictions, identifying a new channel through which geography matters. Third, we quantify how these geographic channels affect interest rate dispersion, the international transmission of financial shocks, and the allocation of capital and the resulting distribution of output.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 documents the geographic relationship in bilateral interest rate differentials and its persistent component. Section 4 develops the model and characterizes the equilibrium interest rate responses to destination attractiveness. Section 5 estimates bilateral portfolio frictions and the elasticity of portfolio demand to returns. Section 6 estimates and recovers the spatial component of destination attractiveness. Section 7 presents the quantitative decomposition, international shock-transmission analysis, and steady state counterfactuals. Section 8 concludes.

Related Literatures Our paper relates to several strands of the literature. First, it contributes to the open-economy macroeconomics literature studying international capital allocation and global portfolio positions, including Lucas (1990), Obstfeld and Rogoff (1995), Gourinchas and Jeanne (2006), Pavlova and Rigobon (2007), Gourinchas and Rey (2007), Caselli and Feyrer (2007), Alfaro et al. (2008), Pellegrino et al. (2025), and Kleinman et al. (2026). Our analysis is also related to

models of segmented international financial markets, including Gabaix and Maggiori (2015), Maggiori (2017), and Itskhoki and Mukhin (2021, 2025), which emphasize frictions that prevent capital from being allocated frictionlessly across countries.

Second, our analysis relates to the literature documenting gravity-type patterns in international capital allocation. Portes et al. (2001) and Portes and Rey (2005) show that international financial transactions decline with geographic distance, with related evidence provided by Aviat and Coeurdacier (2007) and Lane and Milesi-Ferretti (2008). Coeurdacier and Martin (2009) and Okawa and van Wincoop (2012) incorporate gravity-type portfolio relationships into models of international asset allocation. More recently, Koijen and Yogo (2026) exploit gravity-type variation in asset demand to construct instruments for estimating the sensitivity of international portfolio demand to returns. Our paper builds on this literature by treating geographic distance not only as a bilateral portfolio friction but also as a source of spatial structure in destination-side investment attractiveness.

Third, the paper relates to work studying spatial dependence in macroeconomics and finance, including LeSage and Llano (2013), Dell'Erba et al. (2013), Lustig and Richmond (2020), Servén and Abate (2020), Krisztin and Piribauer (2021), and Shambaugh and Zhou (2024). This literature highlights that macroeconomic and financial outcomes can exhibit systematic dependence across geographically connected economies. Our contribution is to bring this spatial perspective into a model of international portfolio allocation by allowing destination-specific attractiveness to contain a spatially correlated persistent component.

In terms of modeling, our framework is related to dynamic spatial equilibrium models, including Allen and Arkolakis (2025) and Eckert and Kleineberg (2024), as well as the asset demand system developed by Koijen and Yogo (2019, 2026). More specifically, our model is closely related to Kleinman et al. (2026) and Pellegrino et al. (2025), who embed international asset demand into open-economy general equilibrium models to study the allocation of capital across countries. Pellegrino et al. (2025), in particular, study how bilateral geographic frictions, taxation, and political risk shape global capital allocation. Our paper instead studies how geography shapes international capital allocation through two distinct channels: bilateral investor–destination portfolio frictions and spatially correlated destination-specific heterogeneity. We then quantify the contribution of each channel to cross-country interest rate dispersion.

2 Data

Interest Rates and Inflation We obtain interest rate data from the Global Macro Database (Müller et al., 2025). To measure returns on capital while minimizing the influence of short-run monetary policy, we use long-term interest rates. For countries with incomplete coverage, we supplement the series with 10-year government bond yields from the Covered Interest Rate Parity Deviations dataset (Du and Schreger, 2016; Du et al., 2018, 2025). We construct real interest rates using five-year-ahead inflation expectations from the IMF World Economic Outlook Historical Forecasts Database (International Monetary Fund, 2026b).

Bilateral Capital Holdings We construct bilateral portfolio holdings as the sum of short- and long-term debt securities and equity securities. Bilateral holdings are obtained from the IMF Coordinated Portfolio Investment Survey (CPIS) (International Monetary Fund, 2026a). To construct total amounts outstanding, we combine information from the OECD, BIS Debt Securities Statistics (Bank for International Settlements, 2026), and World Bank market capitalization data (World Bank, 2026). For the United States, we supplement the CPIS data with Treasury International Capital (TIC) data (U.S. Department of the Treasury, 2026). To address distortions arising from financial centers and tax havens, we reallocate portfolio positions from a residency basis to an ultimate nationality basis using the Global Capital Allocation Project of Coppola et al. (2021). The implementation follows the data construction in the replication package of Koijen and Yogo (2026).

Country Characteristics We compile country-level and bilateral characteristics from several sources. Sovereign credit default swap (CDS) spreads and sovereign credit rating scores are obtained from Refinitiv Datastream. Capital-account openness is measured using the Chinn–Ito index (Chinn and Ito, 2006), and political stability is obtained from the World Bank Worldwide Governance Indicators (Kaufmann et al., 2011). To capture geopolitical alignment, we use United Nations General Assembly ideal-point estimates of Bailey et al. (2017), based on the voting data of Voeten et al. (2009), and geopolitical bloc classifications from the replication package of Gopinath et al. (2025). Bilateral geographic distance and other country-pair characteristics are obtained from the CEPII GeoDist database (Mayer and Zignago, 2011). Population data are obtained from the World Bank World Development Indicators (World Bank, 2026).

3 Motivating Empirical Evidence

This section investigates the spatial relationship of interest rates across countries. If frictions to capital movement or other relevant factors exhibit geographic patterns, interest rates should also exhibit systematic geographic patterns. In particular, geographically closer countries may have more similar interest rates, with bilateral interest rate differentials increasing with bilateral distance. To examine this relationship, we consider two empirical specifications. The first uses a pooled regression, while the second isolates the time-invariant component.

3.1 Geographic Relationship in Interest Rate Differentials

We consider the following regression.

$$|r_{i,t} - r_{j,t}| = \beta \ln(dist_{i,j}) + X_{i,j}\Gamma + \alpha_i + \alpha_j + \lambda_t + \varepsilon_{ijt}$$

For each country pair, the interest rate differential is measured by the absolute value of the difference in interest rates. Interest rates are measured in percentage points; a one-unit difference corresponds to a one-percentage-point interest rate gap.. The vector $X_{i,j}$ contains bilateral controls, including indicators for whether the two countries share a border or a common language, have a colonial relationship, or were historically part of the same country.

The coefficient β measures how interest rate differences vary with geographic distance. A positive value indicates that differences increase as the distance between two countries increases, suggesting that geographically closer countries are more likely to have similar interest rates. The two country fixed effects absorb systematic differences driven by each country's overall interest rate position. Intuitively, country fixed effects allow us to compare how interest rate differences vary across a given country's counterparts. For example, conditional on the U.S. fixed effect, the regression compares whether interest rate differentials tend to be larger with geographically more distant countries.

Panel A in Table 1 shows that geographically more distant country pairs tend to exhibit larger interest rate differentials. The coefficient on geographic distance remains positive and statistically significant after controlling for bilateral characteristics, including shared borders, common language, colonial ties, and historical state relationships. The result shows that geographically proximate countries tend to have similar interest rates, with the similarity decreasing as distance increases.

3.2 The Persistent Component of Interest Rate Differentials

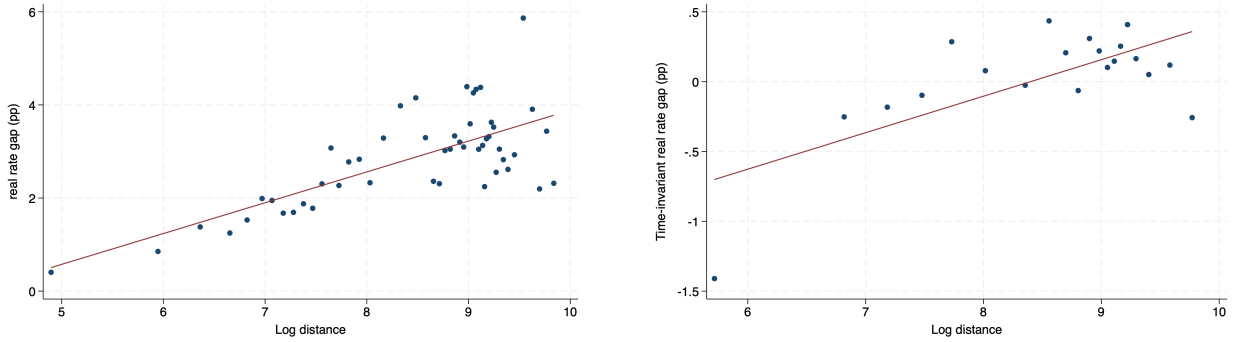
The specification in Section 3.1 shows that interest rate differentials are larger as the distance between two countries increases. However, this relationship may partly reflect transitory shocks. For example, geographically neighboring countries may face similar transitory monetary or macroeconomic shocks and therefore exhibit similar interest rates. We therefore investigate whether geographic distance is also related to the persistent component of bilateral differentials, suggesting a systematic geographic pattern in interest rates. We first decompose bilateral interest rate differentials into two country-year components and a time-invariant country-pair component, $\delta_{i,j}$. The country-year fixed effects absorb country-specific time-varying shocks.

$$|r_{i,t} - r_{j,t}| = \alpha_{i,t} + \alpha_{j,t} + \underbrace{\delta_{i,j}}_{\text{pair FE}} + u_{i,j,t}$$

In the second stage, we regress the estimated pair component on bilateral geographic distance with the same bilateral controls used in the baseline specification.

$$\hat{\delta}_{i,j} = \beta^P \ln(dist_{i,j}) + X_{i,j}\Gamma + \eta_{i,j}$$

Panel B in Table 1 reports the regression estimates, and Figure 2 (b) shows a binscatter plot of the δ component against log distance. The estimates and figure show that geography plays a systematic role in bilateral interest rate differences.



(a) Pooled panel

(b) Persistent component

Figure 2: Interest-Rate Differentials and Geographic Distance

Table 1: Geographic Distance and Interest-Rate Differentials

Panel A. Pooled Panel

	(1)	(2)	(3)	(4)
Log distance	0.232***	0.228***	0.242***	0.236***
	(0.039)	(0.047)	(0.036)	(0.045)
Shares border		-0.036		-0.109
		(0.120)		(0.128)
Common language			0.104	0.076
			(0.096)	(0.105)
Colonial relationship				0.208*
				(0.109)
Same country (past/now)				0.121
				(0.160)
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	56088	56088	56088	56088
Adj. R-squared	0.504	0.504	0.504	0.504

Panel B. Persistent Pair Component

	(1)	(2)	(3)	(4)
Log distance	0.261***	0.275***	0.261***	0.283***
	(0.023)	(0.025)	(0.023)	(0.025)
Shares border		0.226		0.101
		(0.146)		(0.155)
Common language			-0.004	-0.082
			(0.079)	(0.082)
Colonial relationship				0.447***
				(0.158)
Same country (past/now)				0.354
				(0.244)
Observations	4323	4323	4323	4323
Adj. R-squared	0.028	0.029	0.028	0.030

4 Model

In this section, we develop a dynamic multi-country production economy. There are N countries, indexed by $n \in \{1, \ldots, N\}$. In the economy, each country is populated by representative households and competitive firms. Households choose consumption and saving, which financial intermediaries allocate across countries. International investment is subject to bilateral frictions, which make identical investment opportunities valued differently depending on the investor's location. This setup aligns with recent literature on segmented capital markets (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021, 2025) and is closely related to international portfolio allocation models in Pellegrino et al. (2025) and Kleinman et al. (2026).

4.1 Households

The representative household in country n is endowed with a fixed amount of labor, ℓ_n. The household chooses consumption and saving to maximize lifetime utility,

$$\max_{\{c_{nt}, a_{n,t+1}\}} \sum_{t=0}^{\infty} \beta_n^t \, u(c_{nt}) \tag{1}$$

$$\text{subject to } c_{n,t} + a_{n,t+1} = \mathcal{R}_{n,t} a_{n,t} + w_{n,t} \ell_n$$

where β_n is the country-specific discount factor,[1] and $\mathcal{R}_{n,t}$ denotes the return on the household's savings portfolio. The optimal consumption and saving decision satisfies $u'(c_{n,t}) = \beta_n \mathcal{R}_{n,t+1} u'(c_{n,t+1})$. In steady state, consumption and portfolio returns are constant over time, so the Euler equation reduces to $1 = \beta_n \mathcal{R}_n$.

4.2 International Capital Allocation

Financial intermediaries allocate household savings across countries. An investor in country n considering investment in country i faces a bilateral portfolio wedge Φ_{ni}.

The wedge captures frictions that discourage investment in a particular foreign country. It is investor-destination pair specific, so even for the same investment destination, frictions may differ by investor country. We model the wedge as a function of geographic distance and geopolitical

[1] Cross-country differences in discount factors may reflect differences in mortality risk, as in Pellegrino et al. (2025), or differences in time preferences, as documented by Falk et al. (2018).

alignment. First, we assume that the wedge increases with bilateral geographic distance. This reflects information frictions. Investors from farther away face greater difficulty acquiring information about the destination market because of limited familiarity or physical inaccessibility. Second, we allow the wedge to widen when investors invest in countries belonging to different political blocs. Recent literature on geoeconomic fragmentation documents that cross-border FDI and portfolio holdings tend to be lower between geopolitically misaligned countries (e.g., Gopinath et al., 2025). Therefore, the wedge captures both geographic and geopolitical barriers to international capital allocation.

The ω term captures investment attractiveness other than the interest rate. This component is common to investors from all countries of origin. While the bilateral portfolio wedge ϕ varies across investors, ω depends only on destination country i and time t. It therefore captures characteristics of country i and how global investors view the country, both of which affect portfolio allocation beyond the interest rate. A higher ω_{it} increases the portfolio share allocated to country i by investors from all countries. One potential source of this is country credit risk. Countries with lower credit risk may be more attractive to international investors.

For each unit of wealth, the intermediary draws an idiosyncratic taste shock ε_{ni} for each destination i, independently across destinations, from a Gumbel distribution.

$$F(x) = \exp(-\exp(-x + \bar{\gamma}))$$

where $\bar{\gamma}$ is Euler–Mascheroni constant. The scale of the idiosyncratic shock is governed by σ_N. The intermediary in country n invests in country i if the friction- and preference-adjusted return from investing in i is the highest among all available destinations.

$$\varepsilon(n,i,\nu)\sigma_N - \Phi_{ni} + \omega_{it} + r_{i,t} = \max_h \{\varepsilon(n,i,\nu)\sigma_N - \Phi_{nh} + \omega_{ht} + r_{h,t}\} \tag{2}$$

The share of country n's savings allocated to country i through financial intermediaries is given by

$$b_{nit} = \frac{a_{nit}}{a_{nt}} = \frac{\exp(-\Phi_{ni} + \omega_i + r_{i,t})^{1/\sigma_N}}{\sum_h \exp(-\Phi_{nh} + \omega_h + r_{h,t})^{1/\sigma_N}} \tag{3}$$

Here, a_{nit} denotes the amount of country n's savings allocated to country i, while a_{nt} denotes total savings in country n. A larger portfolio wedge Φ_{ni} reduces the share allocated to destination i, whereas a higher destination-specific attractiveness ω_{it} or a higher interest rate r_{it} increases its portfolio share. The parameter σ_N governs the dispersion of idiosyncratic investment preferences

and, therefore, the sensitivity of portfolio allocation to returns and destination attractiveness. Then the resulting gross return on the portfolio is the share-weighted average of returns:

$$\mathcal{R}_{n,t} = 1 + \sum_i b_{nit} r_{i,t} \tag{4}$$

4.3 Firm

Each country n has a representative firm that produces homogeneous goods with Cobb-Douglas production technology. The firm takes the wage $w_{n,t}$ and the interest rate $r_{n,t}$ as given and chooses its factors to maximize profits. Output is produced using capital $k_{n,t}$ and labor $\ell_{n,t}$, with country-specific productivity $z_{n,t}$.

$$r_{n,t+1} = \alpha z_{n,t} k_{n,t}^{\alpha-1} \ell_{n,t}^{1-\alpha}$$

Thus, the interest rate in each country is decreasing in the amount of capital allocated to that country. An increase in capital inflows raises $k_{n,t}$, lowers the marginal product of capital, and thereby reduces the equilibrium interest rate.

4.4 Market Equilibrium

A competitive equilibrium consists of allocations $\{a_{nt}, b_{nit}, k_{nt}\}_{n,i,t}$ and prices $\{\mathcal{R}_{nt}, r_{nt}\}_{n,t}$ such that households optimize given prices and, for all n and t,

$$r_{nt} = \alpha z_n k_{nt}^{\alpha-1}, \tag{5}$$

$$k_{nt} = \sum_i b_{int} a_{it}, \tag{6}$$

$$\mathcal{R}_{nt} = 1 + \sum_i b_{nit} r_{it}. \tag{7}$$

4.5 Destination Attractiveness and Equilibrium Interest Rates

We show how destination attractiveness affects equilibrium interest rates in a general equilibrium. Consider an equilibrium in which investor asset stocks $\{A_n\}$ are fixed, while portfolio shares, destination capital stocks, and interest rates adjust endogenously.

Proposition (Destination Attractiveness and Interest Rates)**.** Consider an equilibrium with strictly positive portfolio shares and decreasing marginal return to capital. The equilibrium response of

interest rates to changes in destination attractiveness is given by

$$d\mathbf{r} = -P\, d\boldsymbol{\omega}, \qquad P = (H^{-1} + J)^{-1} J, \tag{8}$$

where P is the equilibrium transmission matrix, and H and J characterize the responsiveness of capital supply and portfolio demand to interest rates, respectively.

The proof is provided in Appendix I.1.

An increase in destination attractiveness attracts capital inflows, increasing the capital stock and lowering interest rates. In a multi-country open economy, interest rate movements depend on equilibrium portfolio allocation. This proposition shows how a destination-attractiveness shock can lead to interest rate movements in the global economy.

We next examine how a geographically related shock affects interest rates in a general equilibrium. Using the proposition above, we compare two types of attractiveness shocks originating in country i.

Idiosyncratic Shock. Consider a positive attractiveness shock of magnitude $h > 0$ that affects only country i. The shock and its equilibrium interest rate response in country j are given by

$$d\boldsymbol{\omega}_i^I = he_i, \qquad \frac{\partial r_j^I}{\partial h} = -P_{ji}, \tag{9}$$

where e_i is the i-th unit vector.

Under the equilibrium conditions, we have $0 < P_{ii} < 1$ and $P_{ji} < 0$ for $j \neq i$. Therefore a positive idiosyncratic attractiveness shock lowers the interest rate in the origin country while raising interest rates in other countries through international capital reallocation.

Spatially Correlated Shock. Next, we consider a shock originating in country i that has a spatial structure. Let K denote a geographic kernel satisfying $K_{ii} = 1$, with K_{ji} declining with the distance between countries i and j. Then the interest rate response to the shock is given by

$$d\boldsymbol{\omega}_i^S = hKe_i, \qquad \frac{\partial \mathbf{r}^S}{\partial h} = -PKe_i. \tag{10}$$

For a country $j \neq i$,

$$\frac{\partial r_j^S}{\partial h} = \underbrace{-P_{ji}}_{\text{Origin spillover}} - \underbrace{P_{jj}K_{ji}}_{\text{Own attractiveness}} - \underbrace{\sum_{k \neq i,j} P_{jk}K_{ki}}_{\text{Other-country spillovers}} . \tag{11}$$

Compared to an idiosyncratic shock, a spatially correlated shock can produce positive or negative interest rate responses in neighboring countries. Specifically, the interest rate in country j declines if

$$P_{jj}K_{ji} > -P_{ji} - \sum_{k \neq i,j} P_{jk}K_{ki}. \tag{12}$$

This shows that a destination attractiveness shock with a spatial structure can generate macroeconomic responses within a region that differ from those in distant countries.

5 Parameterization

5.1 Estimation of Portfolio Allocation Components

In this section, we estimate the components of portfolio allocation. From equation (3), the portfolio share can be decomposed into three components. The first is the bilateral portfolio friction ($\phi_{ni} = \Phi_{ni}/\sigma_N$). We parameterize this component as a function of geographic distance, home bias, and geopolitical alignment, measured by whether the two countries belong to different geopolitical blocs.[2]

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta^{foreign} \mathbb{1}[n \neq i] + \beta^{crossblock} Crossblock$$

The coefficient β^{dist} captures the friction associated with geographic distance. The coefficient $\beta^{foreign}$ captures home bias, and $\beta^{crossblock}$ captures the additional friction associated with investing in countries belonging to different geopolitical blocs. Second, the destination-specific component, $(\omega_{it} + r_{it})/\sigma_N$, is common across investors and is absorbed by destination-country $\times$ time fixed effects. Third, the denominator is absorbed by investor-country $\times$ time fixed effects.

[2]We use the classification of Gopinath et al. (2025). Countries are classified into three groups—the U.S. bloc, the China bloc, and non-aligned countries—based on their ideal points in the United Nations General Assembly.

$$b_{nit} = \underbrace{\frac{\overbrace{\exp(-\phi_{ni})}^{\text{Capital Moving Friction}} \overbrace{\exp(\omega_{it}/\sigma_N + r_{it}/\sigma_N)}^{\text{Time}\times\text{Destin FE}}}{\sum_h \exp(-\phi_{nh} + \omega_{ht}/\sigma_N + r_{ht}/\sigma_N))}}_{\text{Time}\times\text{Origin FE}} \tag{13}$$

Estimation Results. The estimated β^{dist} is -0.797. The estimated $\beta^{foreign}$ is -4.159, suggesting substantial home bias in portfolio allocation, consistent with the literature. The estimated $\beta^{crossblock}$is -0.796. Appendix A.1 reports the detailed estimation results and robustness checks.

5.2 Portfolio Demand Elasticity

To recover ω, we need to estimate σ_N. However, as interest rates are determined in equilibrium, estimating this parameter requires an instrumental variable strategy.

Building on the approach of Koijen and Yogo (2026), we construct an instrument based on predicted asset demand relative to predicted asset supply. We use the estimated gravity relationships to predict investment demand from third-party countries. Then, combining this with supply predicted from observable characteristics, we construct relative demand pressure. Finally, we measure this pressure relative to that of the investor's home country. The resulting measure of residual demand pressure generates variation in equilibrium interest rates through market clearing.

Using this instrument, we estimate the sensitivity of portfolio allocation to interest rates by two-stage least squares. The estimated coefficient is 13.58, implying $\hat{\sigma}_N = 0.074$. Appendix A.2 provides the details of the instrument construction, identification assumptions, and estimation results.

5.3 Steady-State Calibration

The remaining parameters are calibrated to match the baseline steady state for each country. We choose β_n to satisfy the steady-state Euler equation and calibrate z_n to match the observed interest rate.

6 Spatial Structure and Origins of Destination Attractiveness

6.1 Geographic Patterns in Destination Attractiveness

In the model, ω_{it} captures destination attractiveness to global investors. If these components are spatially correlated, geographically proximate countries should have similar values, which may generate regional clusters in capital allocation. This provides a distinct channel that affects international capital allocation, separate from bilateral portfolio frictions. While bilateral frictions operate between an investor and a destination, spatially correlated components of ω operate at the destination-country level, independent of the investor's location. For example, a U.S. investor may allocate less capital to Korea than to Canada because Korea is geographically more distant from the United States. This bilateral channel, however, does not capture the fact that Korea may have a demand component similar to that of nearby countries such as Japan. If destination attractiveness is spatially correlated, investors from all origins may view Korea and Japan as more similar than geographically distant countries such as Argentina. Therefore, the spatial structure of destination attractiveness captures these geographic relationships across destinations. To test this relationship, we consider the following regression:

$$|\omega_{it} - \omega_{jt}| = \beta^{\omega} \ln(dist_{ij}) + X'_{ij}\Gamma + \alpha_i + \alpha_j + \lambda_t + \varepsilon_{ijt}.$$

A positive coefficient on distance indicates that geographically distant countries tend to have more dissimilar destination components. Table 2a shows a positive and statistically significant relationship between bilateral distance and differences in ω_{it}. The relationship is robust to controls for contiguity, common language, colonial ties, and historical state relationships.

Table 2a shows a positive relationship between geographic distance and differences in ω_{it}. A positive β^{ω} indicates that geographically distant countries tend to have more dissimilar destination components. Because asset holdings partly reflect market size, we remove this effect by residualizing with respect to population and use the resulting components in our baseline analysis.

$$u_{it} = \omega_{it} - X'_{it}\hat{\Gamma} - \hat{\lambda}_t, \tag{14}$$

where X_{it} includes log population. We then estimate the analogous regression

$$|u_{it} - u_{jt}| = \beta^u \ln(dist_{ij}) + Z'_{ij}\Pi + \alpha_i + \alpha_j + \lambda_t + \nu_{ijt}.$$

Table 2b shows that the relationship remains. Geopolitical distance also significantly relates to differences in destination attractiveness, and controlling for it reduces the coefficient on geographic distance.

(a) Distance Regression on ω differences

	(1)	(2)	(3)	(4)
ln(distance)	0.0232***	0.0252***	0.0245***	0.0262***
	(0.0037)	(0.0041)	(0.0038)	(0.0042)
Contiguity		0.0237***		0.0138
		(0.0090)		(0.0090)
Common language			0.0263***	0.0238***
			(0.0072)	(0.0072)
Colonial link				0.0010
				(0.0105)
Same country				0.0193*
				(0.0102)
Country 1 FE	Yes	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes	Yes
Year FE	YES	YES	YES	YES
Observations	57,100	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324	4,324
Adj. R-squared	0.580	0.581	0.581	0.582

(b) Distance Regression on u differences

	(1)	(2)	(3)
ln(distance)	0.0263***		0.0099***
	(0.0015)		(0.0016)
Ideal-point distance		0.0620***	0.0551***
		(0.0028)	(0.0031)
Country 1 FE	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324
Adj. R-squared	0.042	0.099	0.103

Table 2: Spatial relationships of destination attractiveness

Table 2b shows that the relationship remains. Geopolitical distance also significantly relates to differences in destination attractiveness, and controlling for it reduces the coefficient on geographic distance. Nevertheless, geographic distance remains important, suggesting that both geographic and geopolitical proximity matter for similarities in destination attractiveness.

6.2 Spatial Decomposition of Destination Attractiveness

Spatial Covariance Structure. We decompose the residual component u_{it} into a persistent spatial component, s_i, a persistent non-spatial component, a_i, and a transitory component, ε_{it}:

$$u_{it} = \underbrace{s_i}_{\substack{\text{Spatial} \\ \text{Component}}} + \underbrace{a_i}_{\substack{\text{Non-spatial} \\ \text{Component}}} + \varepsilon_{it}.$$

As in a random effects framework, we assume that the components are Gaussian

$$s \sim N(0, \Sigma_s), \qquad a \sim N(0, \sigma_a^2 I), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2).$$

To capture the spatial structure, we parameterize the spatial covariance matrix using a Gaussian covariance function.[3]

$$[\Sigma_s]_{ij} = \sigma_s^2 \exp\left(-\left(\frac{d_{ij}}{\rho}\right)^2\right)$$

, where σ_s is scale parameter of spatial components while ρ is decay parameter which measures how quickly spatial dependence decay over distance.

Identification and Estimation. To estimate the parameters governing the structure of each component, we assume that the spatial component, the persistent non-spatial component, and the transitory component are mutually orthogonal. In addition, ε_{it} is uncorrelated across countries and over time. Averaging over time gives

$$\bar{u}_i = \frac{1}{T_i}\sum_t u_{it} = s_i + a_i + \bar{\varepsilon}_i.$$

The parameters σ_s^2 and ρ are identified from cross-country covariance. For two distinct countries $i \neq j$,

$$\mathrm{Cov}(\bar{u}_i, \bar{u}_j) = \sigma_s^2 \exp\left[-\left(\frac{d_{ij}}{\rho}\right)^2\right],$$

where d_{ij} denotes chordal distance.[4] Under the orthogonality assumptions, cross-country covariance reflects only the spatial component. Its magnitude and variation with bilateral distance provide identifying information about σ_s^2 and ρ. For the same country observed in two different periods,

$$\mathrm{Cov}(u_{it}, u_{i\tau}) = \sigma_s^2 + \sigma_a^2, \qquad t \neq \tau.$$

Given the spatial variance, within-country persistence identifies σ_a^2. The unconditional variance is given by

$$\mathrm{Var}(u_{it}) = \sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2.$$

The remaining variance after accounting for the two persistent components identifies σ_ε^2.

We jointly estimate $(\sigma_s^2, \rho, \sigma_a^2, \sigma_\varepsilon^2)$ by restricted maximum likelihood (REML), controlling for log

[3]The Gaussian covariance function, also known as the squared-exponential covariance function, can be obtained as the limiting case of the Matérn covariance family as the smoothness parameter tends to infinity.

[4]Chordal distance is the Euclidean distance between two points on the Earth's surface embedded in three-dimensional space. Using this distance ensures that the Gaussian covariance kernel is positive semidefinite.

population and year fixed effects. The estimation uses 1,028 country-year observations from 67 countries over 2003–2020. Table 3 reports the estimates. The estimated $\hat{\rho}$ is 2,929 km, suggesting that the correlation of the spatial component falls to 0.5 at a geographic distance of approximately 2,439 km and to 0.1 at approximately 4,445 km. A parametric-bootstrap likelihood-ratio test with 500 replications rejects the null hypothesis $H_0 : \sigma_s^2 = 0$, with an LR statistic of 36.609 and a bootstrap p-value of 0.002.

Table 3: Spatial Decomposition of Destination Attractiveness

Parameter	Estimate	Interpretation
σ_s^2	0.01687	Spatial persistent variance
ρ	2,929 km	Spatial correlation range (chordal)
σ_a^2	0.00816	Non-spatial persistent variance
σ_ε^2	0.00352	Transitory variance
$\sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2$	0.02855	Total variance of u_{it}

Recovery of Latent Components. After estimating the variance components, we recover the spatial, non-spatial, and transitory components from the residuals for each country. The labor economics literature has used empirical Bayes and BLUP approaches to recover latent components from observed outcomes (Kane and Staiger, 2008; Chetty et al., 2014). We apply the same approach using the estimated spatial covariance structure. We express the country-level averages and their covariance matrix in vector form as follows:

$$\bar{u} = s + a + \bar{\varepsilon},$$

$$V_{\bar{u}} = \Sigma_s + \sigma_a^2 I + D_\varepsilon.$$

where

$$D_\varepsilon = \operatorname{diag}\left(\frac{\sigma_\varepsilon^2}{T_1}, \ldots, \frac{\sigma_\varepsilon^2}{T_N}\right).$$

The spatial and non-spatial components can be separately recovered using their conditional expectations given the observed average residuals:

$$\hat{s} = E[s \mid \bar{u}] = \Sigma_s V_{\bar{u}}^{-1} \bar{u},$$

$$\hat{a} = E[a \mid \bar{u}] = \sigma_a^2 I V_{\bar{u}}^{-1} \bar{u}.$$

The spatial components are recovered using the cross-country covariance structure. The remaining persistent country-specific variation is attributed to the non-spatial components.

6.3 Credit Risk and Spatial Dependence in Destination Attractiveness

What makes destination attractiveness spatially correlated? Appendix F shows that destination attractiveness is largely explained by credit risk. Countries with lower credit risk are preferred by investors and have higher destination attractiveness. If credit risk is spatially correlated, it may help explain the spatial relationship in ω.

Using credit rating data, we find evidence of a persistent spatial relationship in country credit risk. This suggests that country credit risk may be one factor underlying ω that contributes to its spatial correlation. Appendix D provides additional empirical evidence and details.

7 Quantitative Exercise

For quantitative analysis, we compute the steady-state equilibrium. We average variables over the 2003-2020 period. The model requires a complete N $\times$ N matrix of bilateral holdings across investor and destination countries. Because not all countries report portfolio positions, we restrict the analysis to 52 countries. The remaining countries are grouped into the Rest of the World (ROW). [5]

7.1 Decomposition of Geographic Channels

In the counterfactual analysis, investor countries adjust their portfolio shares, reallocating their fixed savings across destinations. This changes capital stocks in destination countries and their equilibrium interest rates. Holding savings fixed allows us to quantify the role of geographic channels in capital allocation and interest rate differentials.

We measure population-weighted interest rate dispersion relative to the United States as

$$D_{US}(r) = \sum_{i \neq US} w_i \, |r_i - r_{US}|,$$

where w_i denotes country i's population weight among non-U.S. countries. We report the decomposition for the full sample and separately for advanced economies (AE) and emerging market economies

[5] Mauritius is included in the Rest-of-the-World outside option because of its small population.

(EME).

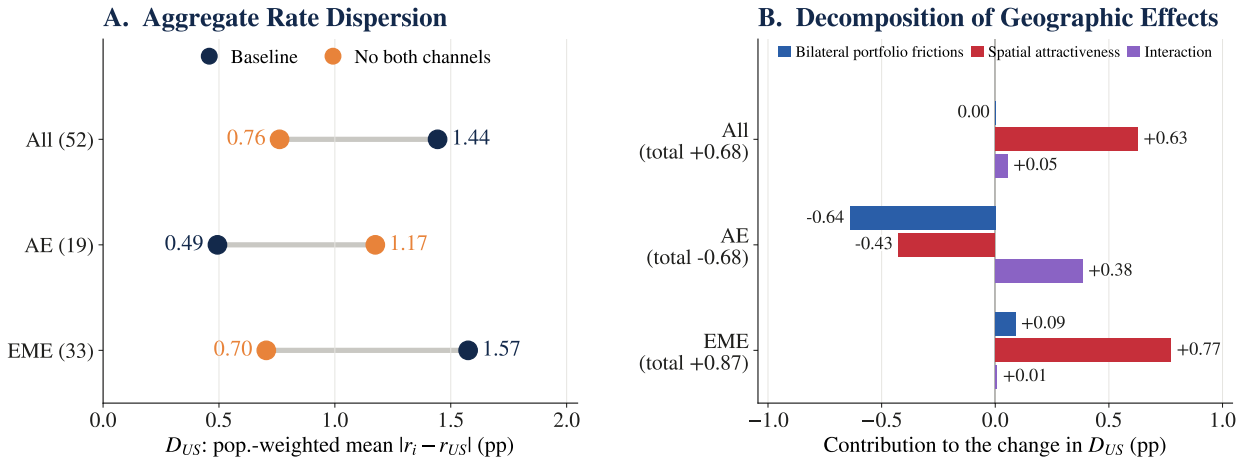


Figure 3: Decomposition Of Interest Rate Differentials Relative to the U.S.

Figure 3 shows the results. In the full sample, interest rate differentials relative to the United States drop from 1.44 to 0.76 when both geographic channels are removed. However, advanced economies experience an increase in the differentials, while emerging economies experience a decline. This shows that the two geographic forces work to reduce interest rate gaps with the U.S. in AEs and widen them in EMEs. In addition, the two groups experience different contributions from the two channels. Bilateral portfolio frictions play a larger role in AEs, while the spatial attractiveness component largely explains interest rate gaps in EMEs.

7.2 Geographic Transmission of Attractiveness Shocks

We show how an idiosyncratic shock and a spatially correlated attractiveness shock introduced in Section 4.5 can differ quantitatively and qualitatively. To compare the outcomes of the two shocks, we consider a one-percentage-point decline in return-equivalent destination attractiveness originating in Peru. An idiosyncratic shock affects only Peru's attractiveness, changing Peru's interest rate while leaving other countries' interest rates unchanged. However, when a negative destination attractiveness shock originating in Peru follows the estimated spatial covariance structure, neighboring countries also experience a decline in attractiveness and an increase in interest rates. Figure 4 plots the interest rate responses to the two shocks against geographic distance from Peru. For the idiosyncratic shock, interest rates decline only slightly in other countries. In contrast,

the spatially correlated shock induces regionally clustered increases in interest rates, while distant countries experience declines.

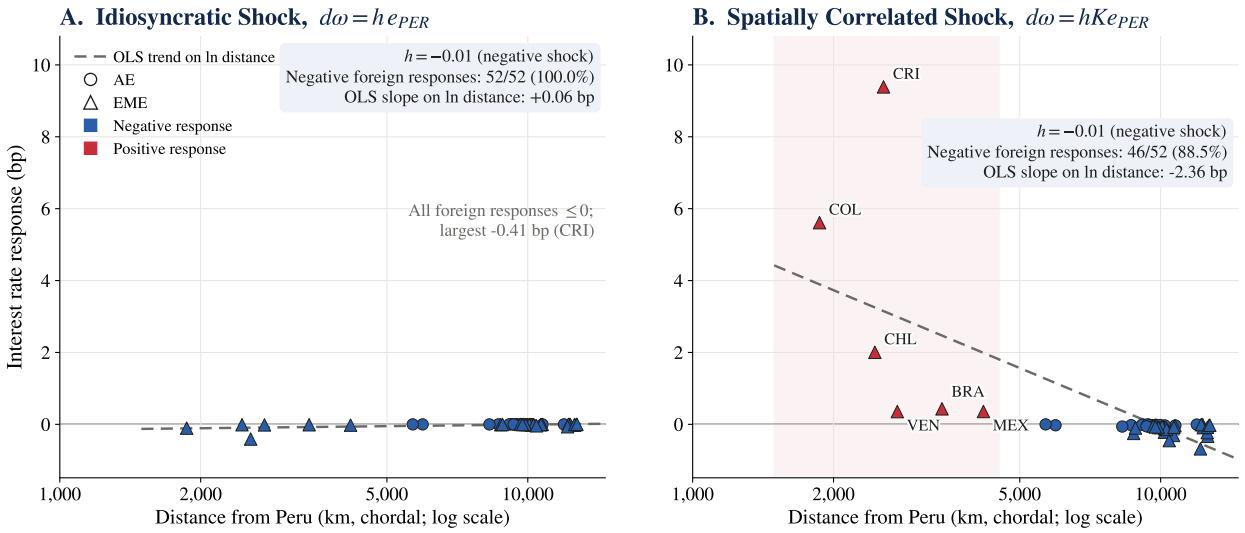


Figure 4: Interest Rate Response to Destination Attractiveness Shocks

7.3 Steady State Counterfactuals

We next solve for the steady-state equilibrium under counterfactual scenarios, allowing saving to adjust to satisfy the household Euler equation. While the preceding analysis holds saving fixed for the decomposition, here we allow saving to vary, so aggregate capital stocks adjust in response to changes in the geographic channels.

Figure 5 compares the variance of log capital and log GDP per capita under the counterfactual scenarios with the baseline. Removing bilateral geographic portfolio frictions reduces the variance of log capital by 25 percent and that of log GDP per capita by 10.8 percent, consistent with Pellegrino et al. (2025). However, the spatially correlated destination attractiveness channel also works in a way that concentrates capital. Removing this component alone reduces the variance of log capital and log GDP per capita by 28.6 percent and 9.9 percent, respectively. When both channels are removed, the reductions become substantially larger, with a 63.8 percent reduction in the variance of log capital and a 24.9 percent reduction in that of log GDP per capita.

These findings suggest that both bilateral frictions and spatially correlated destination attractiveness contribute substantially to global capital concentration and cross-country output inequality.

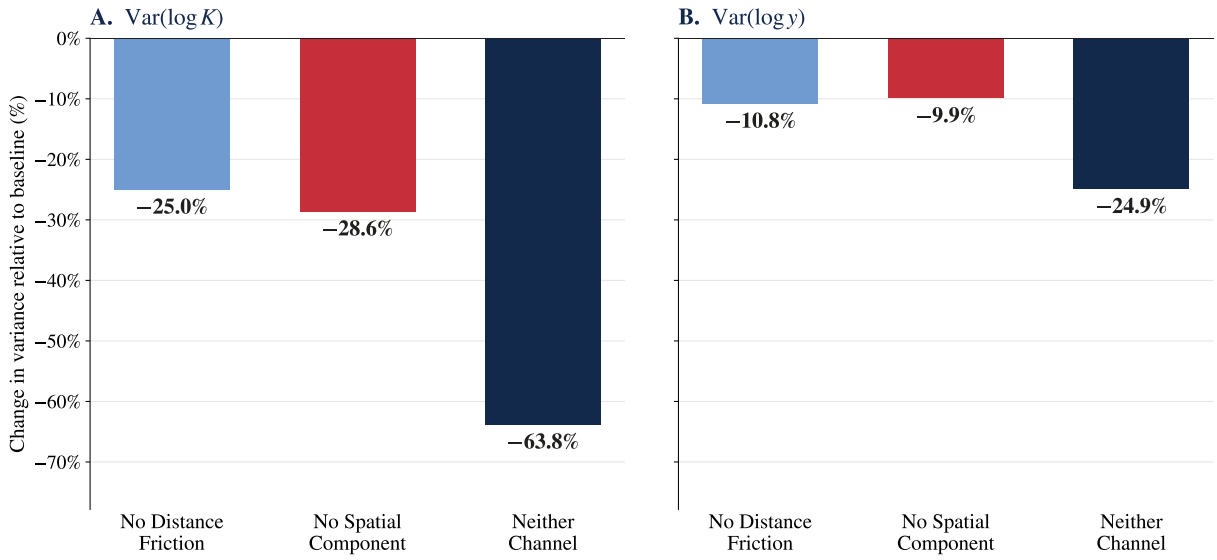


Figure 5: Capital and Output Dispersion Under Counterfactual Scenarios

8 Conclusion

In this paper, we investigate how geography shapes international finance. We document that geographically distant countries tend to have larger interest rate differences, and this relationship persists over time. To explain the empirical evidence, we develop a multicountry general equilibrium model in which two geographic channels affect portfolio allocation. First, geographic distance creates bilateral portfolio frictions. Second, destination attractiveness itself is spatially correlated, providing a distinct geographic channel.

Our quantitative analysis shows that these two distinct geographic channels have important implications. First, geography widens interest rate gaps relative to the U.S. on average, especially for EMEs, while narrowing them for AEs. Second, the spatial structure of destination attractiveness affects how countries respond to shocks. Spatially correlated destination attractiveness shocks generate different interest rate responses in neighboring and distant countries. Third, geography helps explain global capital allocation and output inequality. Removing both geographic channels substantially reduces the variance of log capital and log GDP per capita across countries. These findings suggest that geography matters for international finance not only through bilateral barriers to investment but also through spatial correlation in destination attractiveness.

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